Event-Triggered Adaptive Neural Impedance Control of Robotic Systems

Shuai Ding, Jinzhu Peng, *Member, IEEE*, Hui Zhang, *Member, IEEE*, and Yaonan Wang

***Abstract*— This article presents an event-triggered adaptive neural impedance control (ETANIC) scheme for robotic systems, where the combination of impedance control (IC) and event-triggered mechanism can significantly reduce the computational burden and the communication cost under the premise of ensuring the stability and tracking performances of the robotic systems. The IC is used to achieve the compliant behavior of the robotic systems in response to the environment. The uncertainties of the robotic systems are estimated by the radial basis function neural network (RBFNN), and the update laws for RBFNN are derived from the designed Lyapunov function. The stability of the whole closed-loop control system is analyzed by the Lyapunov theory, and the event-triggered conditions are designed to avoid the Zeno behavior. The numerical simulation and experimental tests demonstrate that the proposed ETANIC scheme can achieve better efficiency for controlling the robotic systems to perform the interaction tasks with the environment in comparison to the adaptive neural IC (ANIC).**

***Index Terms*— Adaptive control, event-triggered control (ETC), impedance control (IC), neural network (NN), robotic system.**

I. Introduction

As ONE of the main compliance control techniques, impedance control (IC) integrates the Cartesian space trajectory and contact force of the robot's end-effector into one framework, which can prevent the problem resulting from the separate control in the orthogonal space of position and force. Therefore, since the concept of IC was first proposed by Hogan [1], IC for robotic manipulator has been widely studied, such as robust IC [2], [3], [4], [5] and hybrid IC [6], [7], [8], [9], [10]. Since the adaptive IC (AIC) does not require the accurate parameter information of the system and environment, which makes the controller design easier, different types of AIC [11], [12], [13], [14], [15], [16], [17], [18] have been proposed for robotic manipulator. Sharifi et al. [12] proposed four model reference adaptive impedance controllers by linearly parameterizing the robotic system. Peng et al. [16] designed an adaptive neural position/force tracking IC strategy for the robotic system, where the neural network (NN)-based adaptive compensator was used to solve the system uncertainties. To realize the target impedance model, Yu et al. [18] used the AIC strategy and a Bayesian scheme to obtain the human impedance parameters and human motion intention recognition. Chien and Huang [19] designed the function approximation technique-based AIC scheme to prevent the computation of the regressor matrix.

Regressor-free control mostly belongs to indirect adaptive control, which can estimate the unknown parameters and the system model online. In [11] and [20], the reinforcement-learning-based impedance controller was proposed for a human–robot cooperation system with transient perturbations or unknown environments. The reinforcement-learning-based control algorithms were also designed to conduct the resilient formation tracking control [21] and the cooperative formation control [22]. Meanwhile, as a popular model-free control method, NN is applied to approximate the nonlinear continuous function and usually used to compensate the nonlinear uncertainties of the robotic system. For example, in the NN-based adaptive force tracking IC scheme [17] and force tracking control strategy [23], the NN was used to deal with uncertain parameters and disturbances. Jung and Hsia [2] presented a robust neural force tracking IC scheme, which used an NN to compensate the uncertainties and achieve a specified force tracking performance. However, for the IC based on the NN [14], [15], [16], [17], [18], [23], [24], linear parameterizing [12], [13], [25], [26], and function approximation technique [19], [27], [28], the weight or parameter matrices in controllers and the system states need to be continuously updated with a constant sampling period. It should be pointed out that most constant period sampling methods would increase the communication between the controller and the robotic systems and will lead to high system cost. Furthermore, in practical applications such as human–robot interactions, it is not necessary for continuous feedback on the current states.

Manuscript received 28 February 2022; revised 14 March 2023; accepted 17 May 2023. Date of publication 31 May 2023; date of current version 8 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62273311, Grant 61773351, Grant 61971071, and Grant 61733004; in part by the National Key Research and Development Program of China under Grant 2018YFB1308200; in part by the Program for Science and Technology Innovation Talents in Universities of Henan Province under Grant 20HASTIT031; and in part by the Key Specialized Research and Development Breakthrough in Henan Province under Grant 222102220117. *(Corresponding author: Jinzhu Peng.)*

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Digital Object Identifier 10.1109/TNNLS.2023.3278301

In recent years, event-based sampling has drawn a substantial concentration of researchers, and various relevant control schemes have been presented [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. Event-triggered control (ETC) reduces the communication of the control-loop system including sensors, controllers, and plants by the definition of the event and triggering mechanisms [30]. Åarzén [29] proposed an event-sampling-based PID controller, which greatly reduced CPU utilization while maintaining control performance. Ngo and Liu [33] presented the ETC-based synchronization control method, which significantly reduced the network burden and computation of the control system. Later, Aström [34] produced an example to compare and analyze the performance of the controller based on event-triggered and periodic sampling. Zhang and Feng [37] investigated output feedback control of linear systems with an event-driven observer, where the continuous- and discrete-time event detectors were considered. The results of [34] and [37] indicated that ETC had better control performance than periodic sampling control under certain conditions.

Although the sampling frequency of the event-based closed-loop control system was reduced, the global uniform ultimate boundedness (UUB) of the whole system could also be guaranteed [37]. In [30], the behavior of the closed-loop control system was approximated to that of a continuous state-feedback system when the parameters of the event generator were properly selected. Moreover, ETC has also been widely used for nonlinear system control problems [40], [41], [42], [43], [44], [45], [46], [47]. Sahoo et al. [40] proposed an approximation-based ETC for uncertain nonlinear systems with an affine form to reduce the utilization of network resources. Narayanan et al. [44] suggested an event-triggered feedback control scheme for the robotic manipulator and determined the event-triggered condition using the Lyapunov stability method. Mu et al. [48] proposed an event-sampled integral reinforcement learning algorithm for the partially unknown nonlinear systems to reduce the samples and transmissions. Besides, the event-triggered mechanism was also integrated into adaptive learning [49] for implementing the adaptive dynamic programming of nonlinear game systems, where the characteristics of dynamic rule and its relationships with the static rules were revealed by the theoretical analysis.

Motived by the above investigations, in this article, an event-triggered adaptive neural IC (ETANIC) scheme is proposed for the robotic systems to reduce the computational burden and communication cost. To the best of the authors' knowledge and investigations, no previous work has been reported on integrating ETC into the compliant control framework for robotic systems. The contributions of this article are summarized as follows.

1) The event-triggered mechanism is introduced into an IC method to facilitate the proposed ETANIC scheme so that the efficiency of computation and communication can be greatly promoted by the event-triggered mechanism, the compliance of robotic systems is achieved by the IC, and the uncertainties and disturbances can be addressed by the adaptive NN.
2) The stability of the closed-loop control system is ensured by analyzing the UUB of state errors in event-triggered interval and weight errors of RBFNN at event-triggered instant. Moreover, the event-triggered conditions are designed for the proposed ETANIC scheme, where the minimal triggered interval is calculated to avoid the Zeno behavior successfully.
3) The proposed ETANIC scheme is verified on a two-rigid-link robotic manipulator and UR5 robot, where the tracking performances and robustness of the robotic system under different model uncertainties and disturbances are analyzed, and the trigger rates under different event-triggered conditions are also discussed. The results show that the proposed ETANIC scheme can achieve position and force tracking with less computational and communication.

The remainder of this article is organized as follows: Section II presents the preliminaries and problem formulation, including the dynamic model of the robot and RBFNN. The ETANIC scheme and the stability analysis of the whole control system are given in Section III. The results and analysis of numerical simulation and experimental tests are conducted in Section IV. The conclusion is demonstrated in Section V.

Notations: In this article, standard notations are used. $\Re$, $\Re^n$, and $\Re^{n\times n}$ represent the real number set, the n-dimensional vector space, and the $n \times n$ real matrix space, respectively. $\|\cdot\|$ denotes the norm of a vector or a matrix. $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum eigenvalue and the maximum eigenvalue of a matrix, respectively. $I_{n\times n}$ and $\mathrm{diag}(\cdot)$ denote the identity matrix and the diagonal function, respectively. $\mathrm{tr}(\cdot)$ represents the trace of a matrix.

II. Preliminaries and Problem Formulation

A. Dynamics of Robotic System

The general dynamic model of an n-rigid-link robotic manipulator can be represented as follows:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau + \tau_e + d_1 \tag{1}$$

where $q, \dot{q}, \ddot{q} \in \Re^n$ denote the position, velocity, and acceleration vectors of joints, respectively; $M(q) \in \Re^{n\times n}$, $C(q,\dot{q}) \in \Re^{n\times n}$, and $G(q) \in \Re^n$ are the inertia matrix, the vector of centripetal and Coriolis force term, and the gravity vector, respectively; $\tau \in \Re^n$ and $\tau_e \in \Re^n$ are the input torque vector and the external torque vector between the robot and the environment, respectively; $d_1 \in \Re^n$ is the vector of unknown bounded disturbances.

Then, the following assumption and properties are given.

Assumption 1: In practical applications, there are inevitably measured errors caused by environmental and payload variations, which make the precise values of the parameter matrices difficult to be obtained. Therefore, the parameters $M(q)$, $C(q,\dot{q})$, and $G(q)$ in (1) are represented by the known nominal parts $M_0(q)$, $C_0(q,\dot{q})$, and $G_0(q)$ and the bounded uncertain parts o_M, o_C, and o_G, respectively,

$$M(q) = M_0(q) + o_M, \quad C(q,\dot{q}) = C_0(q,\dot{q}) + o_C$$
$$G(q) = G_0(q) + o_G.$$

Property 1: The matrix $M_0(q)$ is symmetric, and positive definite, and satisfies

$$M_m\|\varsigma\|^2 \leq \varsigma^{\mathrm{T}} M_0(q)\varsigma \leq M_M\|\varsigma\|^2 \quad \forall \varsigma \in \Re^n$$

where M_m and M_M are positive constants.

Property 2: The skew-symmetric matrix $\dot{M}_0(q) - 2C_0(q, \dot{q})$ can be described as

$$\varsigma^{\mathrm{T}}\left(\dot{M}_0(q) - 2C_0(q, \dot{q})\right)\varsigma = 0 \quad \forall \varsigma \in \Re^n.$$

In the task space, the relationship of positions in the Cartesian space and the joint space of the robotic manipulator can be expressed as the following kinematic mapping:

$$X = P(q) \tag{2}$$

where $X \in \Re^m$ is the position in the Cartesian space, and $P(q)$ is the forward kinematic function.

According to the inverse kinematic and (2), $\dot{q}$ and $\ddot{q}$ can be calculated as follows:

$$\dot{q} = J^{\dagger}(q)\dot{X} \tag{3}$$

$$\ddot{q} = J^{\dagger}(q)\ddot{X} - J^{\dagger}(q)\dot{J}(q)J^{\dagger}(q)\dot{X} \tag{4}$$

where $\dot{X}, \ddot{X} \in \Re^m$ are the velocity and acceleration in the task space, respectively. $J(q) = \partial P(q)/\partial q$ is the Jacobian matrix. $J^{\dagger}(q)$ denotes the pseudoinverse matrix of $J(q)$.

According to Assumption 1 and (2)–(4), the robotic manipulator (1) can be rewritten as

$$M_{X0}\ddot{X} + C_{X0}\dot{X} + G_{X0} + \Delta_1 = \tau + \tau_e + d_1 \tag{5}$$

where $C_{X0} = [C_0(q, \dot{q}) - M_0(q)J^{\dagger}(q)\dot{J}(q)]J^{\dagger}(q)$, $M_{X0} = M_0(q)J^{\dagger}(q)$, $G_{X0} = G_0(q)$, and Δ_1 is the uncertainty of the dynamic system

$$\Delta_1 = o_{MX}\ddot{X} + o_{CX}\dot{X} + o_{GX} \tag{6}$$

where $o_{CX} = [o_C - o_M J^{\dagger}(q)\dot{J}(q)]J^{\dagger}(q)$, $o_{MX} = o_M J^{\dagger}(q)$, and $o_{GX} = o_G$.

Lemma 1: Based on Property 2, the skew-symmetric matrix $\dot{M}_{X0} - 2C_{X0}$ is held and satisfies

$$\varsigma^{\mathrm{T}}(\dot{M}_{X0} - 2C_{X0})\varsigma = 0 \quad \forall \varsigma \in \Re^n.$$

The proof of this Lemma has been given in [17].

B. Neural Network

The uncertainties of dynamic systems are estimated by the following RBFNN with $N_i - N_m - N_o$ structure:

$$y(x) = \sum_{j=1}^{N_m} \xi_j \varphi_j(x) = \Xi^{\mathrm{T}}\Psi(x) \tag{7}$$

where N_i, N_m, and N_o denote the nodes of input layer, hidden layer, and output layer, respectively. $x = [x_1, \ldots, x_{N_i}]^{\mathrm{T}}$ and $y(x) = [y_1(x), \ldots, y_{N_o}(x)]^{\mathrm{T}}$ are the input vector and the output vector, respectively. $\Xi = [\xi_1, \ldots, \xi_{N_m}]^{\mathrm{T}}$ with $\xi_j = [\xi_{j1}, \ldots, \xi_{jN_o}]^{\mathrm{T}}$ being the weight matrix of the hidden layer to the output layer. The Gauss basis function $\Psi(x) = [\varphi_1(x), \ldots, \varphi_{N_m}(x)]^{\mathrm{T}}$ is chosen as the activation function of the hidden layer

$$\varphi_j(x) = e^{-\frac{\|x-\sigma_j\|^2}{\delta_j^2}}, \quad j = 1, 2, \ldots, N_m \tag{8}$$

where σ_j and δ_j denote the center vector and the width of the jth activation function, respectively.

Then, (7) can be used to estimate a continuous unknown function $h(x)$

$$h(x) = \Xi^{*\mathrm{T}}\Psi(x) + \varepsilon_N \tag{9}$$

where Ξ^* is the ideal weight; ε_N is the minimal approximation error.

For the nonlinear function $h(x)$ defined on a compact set Ω and assuming that there exists a compact set Ω_{Ξ}^* that is defined as $\Omega_{\Xi}^* = \{\Xi^* \in \Re^{N_m \times N_o} : \|\Xi^*\| \leq \Xi_m\}$ with $\Xi_m > 0$, the ideal weight matrix Ξ^* can be described as

$$\Xi^* = \arg\min_{\Xi^* \in \Omega_{\Xi}^*} \left\{\sup_{x \in \Omega}\left|h(x) - h(\hat{\Xi}, x)\right|\right\} \tag{10}$$

where $h(\hat{\Xi}, x)$ is the approximation of $h(x)$; $\hat{\Xi}$ is the estimated weight matrix.

Assumption 2 [50]: Without loss of generality, we assume that the minimal approximation error ε_N is bounded such that $\|\varepsilon_N\| \leq D_N$, for $D_N > 0$. And the basis function $\Psi(\cdot)$ is chosen such that $\|\Psi(\cdot)\| \leq \sqrt{D_\Psi}$.

III. Event-Triggered Adaptive Neural Impedance Controller Design and Stability Analysis

The whole task space of the robotic manipulator will be defined according to the constraint in this section, and the reference trajectory input can be obtained. Then, according to the reference trajectory, the procedure of the controller design is given and the stability is analyzed and proved accordingly.

A. Impedance Control

The whole task space of the robotic manipulator is divided into the free space and the contact space. In the free space, the relationship between the position error $E_d = X - X_d$ and the external force F_e of the robot can be expressed as follows:

$$\Lambda_d\ddot{E}_d + D_d\dot{E}_d + K_d E_d = F_e \tag{11}$$

where Λ_d, D_d and K_d are positive constant matrices that are chosen to provide the impedance of the end-effector.

The control objective in the contact space is to guarantee that the external force of the end-effector can track the desired force F_d, i.e., $E_F = F_e - F_d = 0$. Then, the target dynamics can be represented as

$$\Lambda_d\ddot{X} + D_d\dot{X} = F_e - F_d. \tag{12}$$

Remark 1: Considering the difference in control target dynamics, the reference position trajectory X_r in the whole task space can be obtained, where F_e is zero in the free space and will track the desired force F_d in the contact space. According to the reference position trajectory X_r, the required control scheme will be designed to track the desired position trajectory of the whole task space.

According to the dynamic relationship (11) and (12), the reference acceleration of the end-effector can be calculated as follows:

$$\ddot{X}_r = \begin{cases} \Lambda_d^{-1}[F_e - D_d\dot{E}_d - K_d E_d] + \ddot{X}_d, & \text{free space} \\ \Lambda_d^{-1}(-D_d\dot{X} + F_e - F_d), & \text{contact space.} \end{cases} \tag{13}$$

B. Event-Triggered Adaptive Neural Impedance Controller Design

Define a filter error vector as

$$r(t) = \dot{X}(t) - v(t) \tag{14}$$

where $v(t) = \dot{X}_r(t) - \lambda E(t)$, $E(t) = X(t) - X_r(t)$ is the position state error, and λ is a positive matrix.

Then, the robotic system (5) can be replaced by

$$M_{X0}\dot{r}(t) + C_{X0}r(t) + Y_0 - \Delta = \tau(t) \tag{15}$$

where $Y_0 = M_{X0}\dot{v} + C_{X0}v + G_{X0}$, $\Delta = \tau_e(t) + d_1 - \Delta_1$.

According to (9) and (10) and Section II-B, Δ in (15) can be estimated by RBFNN, i.e.,

$$\Delta = \Xi^{*\mathrm{T}}\Psi(\zeta) + \varepsilon_N \tag{16}$$

where $\zeta = (X^{\mathrm{T}}, \dot{X}^{\mathrm{T}}, {\tau_e}^{\mathrm{T}})^{\mathrm{T}}$ is the input vector of RBFNN.

Substituting (16) into (15) yields

$$M_{X0}\dot{r}(t) = \tau(t) - C_{X0}r(t) - Y_0 + \Xi^{*\mathrm{T}}\Psi(\zeta) + \varepsilon_N. \tag{17}$$

Design the adaptive neural IC (ANIC) as

$$\tau(t) = -Kr(t) + Y_0 - \hat{\Xi}^{\mathrm{T}}\Psi(\zeta) \tag{18}$$

where K is a diagonal positive-definite matrix.

Then, the filtered tracking error dynamics can be rewritten by

$$M_{X0}\dot{r}(t) = -(K + C_{X0})r(t) + \tilde{\Xi}^{\mathrm{T}}\Psi(\zeta) + \varepsilon_N \tag{19}$$

where $\tilde{\Xi} = \Xi^* - \hat{\Xi}$ is the weight error of RBFNN.

Remark 2: In [14], [17], and [18], the nominal dynamics of the robot is built by the NN technique so that the controller can avoid the dependence on the nominal model. For most practical applications, the parameters in nominal dynamics can be obtained and used provided that the uncertainties are considered reasonably. In general, more knowledge of the system model considered in the controller design contributes to better control performances. Therefore, by incorporating the system nominal model in the controller design procedure, the ANIC with constant sampling is designed for subsequent development of ETANIC with event-based sampling.

In ANIC, to facilitate the design of the closed-loop control system, the states have to be continuously updated, which need continuous sampling and communication. However, this generally leads to high computation burden and communication cost, which can be improved by the ETC. Therefore, according to the event-based sampling states and ANIC (18), ETANIC can be designed as follows:

$$\tau(t) = -K\bar{r}(t) + \bar{Y}_0(t) - \hat{\Xi}^{\mathrm{T}}\Psi\left(\bar{\zeta}\right), \quad t_k \le t < t_{k+1} \tag{20}$$

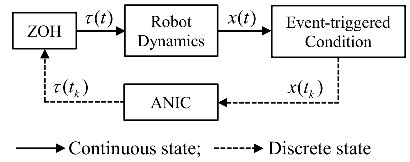


Fig. 1. Structure of the ETANIC system.

with

$$\bar{r}(t) = r(t) - e_r(t), \quad \bar{Y}_0(t) = Y_0(t) - e_Y(t) \tag{21}$$

where $e_r(t)$ and $e_Y(t)$ are the event-based sampling errors of $r(t)$ and $Y_0(t)$, respectively.

Substituting (20) into (17) yields

$$\begin{aligned} M_{X0}\dot{r}(t) = &-(C_{X0} + K)r(t) + Ke_r(t) - e_Y(t) \\ &+ \tilde{\Xi}^{\mathrm{T}}\Psi(\bar{\zeta}) + \varepsilon_N + \Xi^{*\mathrm{T}}\left[\Psi(\zeta) - \Psi(\bar{\zeta})\right]. \end{aligned} \tag{22}$$

Remark 3: In (20), a sequence, $t_k, k = 0, 1, \ldots, \infty$ with $t_0 = 0$, is defined to denote the event-triggered instant. The event-triggered instant t_k is generated entirely by the event-triggered condition. Once the trigger condition is satisfied, the values of state vectors will be sampled, i.e., $x^+(t) = x(t)$ for $t = t_k$. Otherwise, the zero-order hold (ZOH) is used to hold the state vectors, i.e., $\bar{x}(t) = x^+(t_k)$ for $t_k \le t < t_{k+1}$, which will lead to the error $e_x(t) = x(t) - x(t_k) = x(t) - \bar{x}(t)$. Here, $x(t)$ represents the state vectors $r(t)$ and $Y_0(t)$ of the robotic systems. The structure of the ETANIC system is shown in Fig. 1.

C. Stability Analysis

The stabilities of two controllers, ANIC and ETANIC, will be analyzed and proven by the Lyapunov stability theorems.

Theorem 1: Considering the robotic system (1), ANIC is given by (18), and the updated law of NN is designed as follows:

$$\dot{\hat{\Xi}} = \Gamma_1[\Psi(\zeta)r^{\mathrm{T}} - \eta\hat{\Xi}] \tag{23}$$

where the design parameters $\Gamma_1 > 0$ and $\eta > 0$. Then, the tracking errors and adaptive parameter errors are locally UUB.

The proof of Theorem 1 follows a similar approach to that presented in [15] and [17]. Here, the control strategy (18) of the robotic system is given with continuous sampling. Then, the event-based control scheme is designed, and the stability results for the event-based controller will be presented.

Theorem 2: Considering the ETANIC strategy given by (20) for the robotic system (1), the updated law of NN is designed as

$$\hat{\Xi}^+ = (1 - \eta\Gamma_1)\hat{\Xi} + \Gamma_2\Psi(\zeta)r^{\mathrm{T}} \tag{24}$$

where the design parameter $\Gamma_2 = \Gamma_1/(1 + \|r\|^2)$, and event-triggered conditions are set to

$$\|e_r\| \ge \alpha_1\|r\| + \alpha_0, \quad \|e_Y\| \ge \beta_1\|r\| + \beta_0 \tag{25}$$

where $\alpha_1 = \mu/\|K\|$ and $\beta_1 = \mu$ with $\mu \in (0, 1)$, and μ, α_0, and β_0 are known positive constants. Once any trigger conditions in (25) are satisfied, the event will be triggered.

Meanwhile, the proposed ETANIC strategy can guarantee that the tracking and adaptive parameter errors are locally UUB, and the Zeno behavior is also successfully avoided.

Proof: Different from the proof of constant sampled-data control system, the proof of the ETANIC system will be analyzed by the UUB of state errors in the event-triggered interval $t_k < t < t_{k+1}$ and weight errors of RBFNN at the event-triggered instant t_k.

Consider the following Lyapunov function:

$$V_2 = \frac{1}{2} r^{\mathrm{T}} M_{X0} r + \frac{1}{2} \mathrm{tr}\big(\tilde{\Xi}^{\mathrm{T}} \Gamma_2^{-1} \tilde{\Xi}\big). \tag{26}$$

Taking the differentiation of the Lyapunov function (26), we get

$$\dot{V}_2 = \frac{1}{2} r^{\mathrm{T}} \dot{M}_{X0} r + r^{\mathrm{T}} M_{X0} \dot{r} + \mathrm{tr}\big(\tilde{\Xi}^{\mathrm{T}} \Gamma_2^{-1} \dot{\tilde{\Xi}}\big). \tag{27}$$

For $t_k < t < t_{k+1}$, since NN weight is not updated during this time interval, the derivative $\dot{\tilde{\Xi}} = 0$, which means that $\tilde{\Xi}$ is time-invariant. Then, according to Property 2 and using Young's inequality, one can obtain

$$\dot{V}_{21} \le -\lambda_{\min}(K)\|r\|^2 - 2\|r\|^2 + \frac{1}{2}\|K e_r\|^2 + \frac{1}{2}\|e_Y\|^2 + \rho_{210} \tag{28}$$

where $\rho_{210} = (1/2)[2\sqrt{D_\Psi}(\|\tilde{\Xi}\| + \Xi_m) + \|D_N\|^2 + \eta \Xi_m^2]$.

According to the event-triggered conditions (25), we can obtain

$$\dot{V}_{21} \le -(\lambda_{\min}(K) - 2 - \mu^2)\|r\|^2 + \rho_{21} \tag{29}$$

where $\rho_{21} = \rho_{210} + \|K\|^2 \alpha_0^2 + \beta_0^2$.

For $t = t_k$, $r^+ = r$. Hence, the first differentiation of the Lyapunov function (26) can be written as

$$\dot{V}_{22} = \frac{1}{2}\mathrm{tr}\big(\tilde{\Xi}^{+\mathrm{T}} \tilde{\Xi}^{+}\big) - \frac{1}{2}\mathrm{tr}\big(\tilde{\Xi}^{\mathrm{T}} \Gamma_2^{-1} \tilde{\Xi}\big). \tag{30}$$

Using the fact that $\tilde{\Xi}^+ = \Xi^* - \hat{\Xi}^+$, it can be obtained that $\tilde{\Xi}^+ = \tilde{\Xi} - \Gamma_2 \Psi(\zeta) r^{\mathrm{T}} + \eta \Gamma_1 \hat{\Xi}$. Therefore,

$$\dot{V}_{22} = \mathrm{tr}\Bigg(\eta \tilde{\Xi} \Gamma_1 \Gamma_2^{-1} \hat{\Xi} - \eta r \Phi^T(\zeta) \Gamma_1 \hat{\Xi} + \frac{1}{2}\eta^2 \hat{\Xi}^T \Gamma_1^T \Gamma_2^{-1} \Gamma_1 \hat{\Xi} - \tilde{\Xi}\Phi(\zeta) r^T + \frac{1}{2} r \Phi^T(\zeta) \Gamma_2 \Phi(\zeta) r^T\Bigg). \tag{31}$$

According to Young's inequality and considering the fact that $0 \le (\|r\|/(1+\|r\|^2)) \le 1$ and Assumption 2, we can get

$$\dot{V}_{22} \le -\frac{1}{2}\left(\eta \lambda_{\min}\left(\frac{\Gamma_1}{\Gamma_2}\right) - 1\right)\|\tilde{\Xi}\|^2 + \rho_{22} \tag{32}$$

where $\rho_{22} = (1/2)[\eta \Gamma_1 \Xi_m^2 + (2\Gamma_1^2/\Gamma_2 - 1) D_\Psi^2 + (D_\Psi + \eta \Xi_m)^2]$.

Therefore, it can be concluded that

$$\dot{V}_2 \le \begin{cases} -(\lambda_{\min}(K) - 2 - \mu)\|r\|^2 + \rho_{21}, & t_k < t < t_{k+1} \\ -\dfrac{1}{2}\left(\eta \lambda_{\min}\left(\dfrac{\Gamma_1}{\Gamma_2}\right) - 1\right)\|\tilde{\Xi}\|^2 + \rho_{22}, & t = t_k. \end{cases} \tag{33}$$

Based on the above fact and analysis, (26) can be rewritten as

$$V_2 : \begin{cases} V_{21} = \dfrac{1}{2} r^{\mathrm{T}} M_{X0} r, & t_k < t < t_{k+1} \\ V_{22} = \dfrac{1}{2}\mathrm{tr}(\tilde{\Xi}^{\mathrm{T}} \tilde{\Xi}), & t = t_k. \end{cases} \tag{34}$$

Then, considering that (29), (32), and (34), one can obtain

$$\begin{cases} \dot{V}_{21} \le -\kappa_{21}\|r\|^2 + \rho_{21} = -\kappa_{21} V_{21} + \rho_{21} \\ \dot{V}_{22} \le -\kappa_{22}\|\tilde{\Xi}\|^2 + \rho_{22} = -\kappa_{22} V_{22} + \rho_{22} \end{cases} \tag{35}$$

where $\kappa_{21} = ((\lambda_{\min}(K) - 2 - \mu)/(\lambda_{\min}(M_{X0})))$, and $\kappa_{22} = (1/2)(\eta \lambda_{\min}(\Gamma_1/\Gamma_2) - 1)$.

Considering (34) and solving (35), we have

$$\begin{cases} 0 \le V_{21}(t) \le \left(V_{21}(0) - \dfrac{\rho_{21}}{\kappa_{21}}\right) e^{-\kappa_{21} t} + \dfrac{\rho_{21}}{\kappa_{21}} \\ 0 \le V_{22}(t) \le \left(V_{22}(0) - \dfrac{\rho_{22}}{\kappa_{22}}\right) e^{-\kappa_{22} t} + \dfrac{\rho_{22}}{\kappa_{22}}. \end{cases} \tag{36}$$

Equation (36) shows that the error signals r and $\tilde{\Xi}$ converge to $\|r\| \le (2\rho_{21}/(\lambda_{\min}(M_{X0})\kappa_{21}))^{1/2}$, and $\|\tilde{\Xi}\| \le (2\rho_{22}/\kappa_{22})^{1/2}$ as $t \to +\infty$. Therefore, it can be concluded that all the error signals are UUB, and the weight estimation errors decrease and converge.

According to (20), we can conclude that τ is a differentiable function and $\dot{\tau}$ is a function consisting of the bounded closed-loop signals [45]. Therefore, there exists a constant $D_{\dot{\tau}} > 0$ such that $\|\dot{\tau}\| \le D_{\dot{\tau}}$. Moreover, the proposed ETANIC strategy with event-triggered conditions (25) will be proved to avoid the Zeno behavior [47], which means that a time constant $t^* > 0$ exists and satisfies $t_{k+1} - t_k \ge t^* > 0$. When $t_k \le t < t_{k+1}$, taking the differentiation of the event-based sampling errors in (21), we can obtain

$$\frac{d}{dt}\|e_r\| = \frac{d}{dt}\left(e_r^{\mathrm{T}} e_r\right)^{\frac{1}{2}} = \frac{e_r^{\mathrm{T}} \dot{e}_r}{\|e_r\|} \le \|\dot{e}_r\| = \|\dot{r}\| \tag{37}$$

$$\frac{d}{dt}\|e_Y\| = \frac{d}{dt}\left(e_Y^{\mathrm{T}} e_Y\right)^{\frac{1}{2}} = \frac{e_Y^{\mathrm{T}} \dot{e}_Y}{\|e_Y\|} \le \|\dot{e}_Y\| = \|\dot{Y}_0\|. \tag{38}$$

From (18) and (22), we have

$$\|\dot{r}\| \le \frac{D_c\|r\| + \alpha_0 + \beta_0 + \|\tilde{\Xi}\| D_\Psi + D_N}{\lambda_{\min}(M_{X0})} \triangleq D_{\dot{r}} \tag{39}$$

$$\|\dot{Y}_0\| \le D_{\dot{\tau}} + \lambda_{\max}(K)\, D_{\dot{r}} \triangleq D_{\dot{Y}} \tag{40}$$

where $D_c = \|C_{X0}\| + \lambda_{\max}(K)(\alpha_1 + 1) + \beta_1$.

Solving (37) and (38) and considering (25), we can get

$$\begin{cases} \displaystyle\int_{t_k}^{t_{k+1}} \frac{d}{dt}\|e_r\| dt = \alpha_1\|r\| + \alpha_0 \le (t_{k+1} - t_k)\,\|\dot{r}\| \\ \displaystyle\int_{t_k}^{t_{k+1}} \frac{d}{dt}\|e_Y\| dt = \beta_1\|r\| + \beta_0 \le (t_{k+1} - t_k)\,\|\dot{Y}_0\|. \end{cases} \tag{41}$$

Then, the lower time bound exists and satisfies

$$t_{k+1} - t_k \ge \min\left(\frac{\alpha_1\|r\| + \alpha_0}{D_{\dot{r}}}, \frac{\beta_1\|r\| + \beta_0}{D_{\dot{Y}}}\right) > 0. \tag{42}$$

Therefore, the lower bound of the time interval of the trigger exists and is greater than zero so that the Zeno behavior can be successfully avoided. This completes the proof.

TABLE I
PARAMETERS OF THE TWO-RIGID-LINK ROBOT

Parameter	m_1	m_2	$l_1 = l_2$	$l_{c1} = l_{c2}$	$I_1 = I_2$
Nominal	1.6kg	0.9kg	0.75m	0.375m	0.023kg·m^2
Actual	1.8kg	1.0kg	0.80m	0.400m	0.025kg·m^2

Remark 4: When the event is triggered ($t = t_k$), the states of the robotic system are held under the event-triggered mechanism, and the controller's output and RBFNN's parameters are updated. When the event-triggered conditions are not satisfied ($t_k < t < t_{k+1}$), the controller's output is held, and the robotic system is controlled by the torque input calculated at the last trigger instant, which can reduce the number of communication and calculation burden. From (25), it can be concluded that during the event-triggered interval, the sampling errors e_r and e_Y have the upper bound to avoid event triggering, and the controller's output is updated to drive sampling errors back to the corresponding bound at the event-triggered instant.

IV. SIMULATIONS AND EXPERIMENTAL TESTS

In this section, the numerical simulation and experimental tests for ANIC and ETANIC are conducted on two virtual robotic systems, respectively, to demonstrate the effectiveness of the proposed ETANIC strategy.

A. Numerical Simulation

In this section, the effectiveness of two controllers, ANIC and ETANIC, is verified by a two-rigid-link robotic manipulator, whose dynamics is described in Appendix A.

The nominal values and actual values of the parameters of the robotic manipulator used in numerical simulation are chosen as shown in Table I. The initial conditions are $q_1(0) = \pi/2$ rad, $q_2(0) = \pi/3$ rad, and $\dot{q}_1(0) = \dot{q}_2(0) = 0$ rad/s. The desired position and velocity trajectory are given by $X_d(t) = [1.2\sin(0.4t - \pi/4), 1.2\cos(0.4t - \pi/4)]^{\mathrm{T}}$ m and $\dot{X}_d(t) = [0.48\cos(0.4t - \pi/4), -0.48\sin(0.4t - \pi/4)]^{\mathrm{T}}$ m/s, respectively. We take the contact force $F_e = [k_e(X_x - X_{xe}), 0]^{\mathrm{T}}$, where $k_e = 5000$ N/m and $X_{xe} = 1$ m. The desired force input is $F_d = [30, 0]^{\mathrm{T}}$ N. The simulation sampling time and total time are 0.001 and 15 s, respectively.

The parameters of the controller are designed as $\lambda = 40I_{2\times2}$, $K = 75I_{2\times2}$, $\mu = 0.5$, $\alpha_0 = 0.15$, and $\beta_0 = 0.25$. The impedance parameters are selected as $\Lambda_d = 2I_{2\times2}$, $D_d = 75I_{2\times2}$, and $K_d = 330I_{2\times2}$. The structure and parameters of RBFNN are set as $N_i = 6$, $N_m = 10$, $N_o = 2$, $\sigma_j = [-3, 3]_{10}$, $\delta_j = 2$, $\Gamma_1 = 0.01$, and $\eta = 10$. To show the performances of ETANIC, we choose the external disturbances $d_1 = [2\cos(0.5t), \sin(0.5t)]^{\mathrm{T}}$.

1) Comparative Simulation Between ANIC and ETANIC: The simulation results of the ANIC (constant sampling) and ETANIC (event-based sampling) are presented in Figs. 2 and 3. The time interval of the trigger is given in Fig. 4(a), where the asterisk symbols depict when a new event occurs, and the vertical lines represent the associated time interval of the trigger. The norms of NN weight are shown in Fig. 4(b). As shown in Figs. 2 and 3, the position and force tracking can be obtained in both the controllers. From Figs. 2–4(a), one can conclude that under the same condition, the proposed ETANIC scheme has similar trajectory tracking performances to ANIC but less computation burden than ANIC, and the time interval of the trigger also indicates that the Zeno behavior can be avoided. These results demonstrate that the proposed ETANIC scheme can improve the control performances of the robotic system compared with ANIC. Fig. 4(b) indicates that the norms of NN weight in ETANIC are bounded. To demonstrate the control performances of ETANIC directly, the root mean square (rms) of tracking errors and trigger rates for different μ are calculated in ETANIC. According to Table II, when $\mu = 0.5$, the trigger rates of

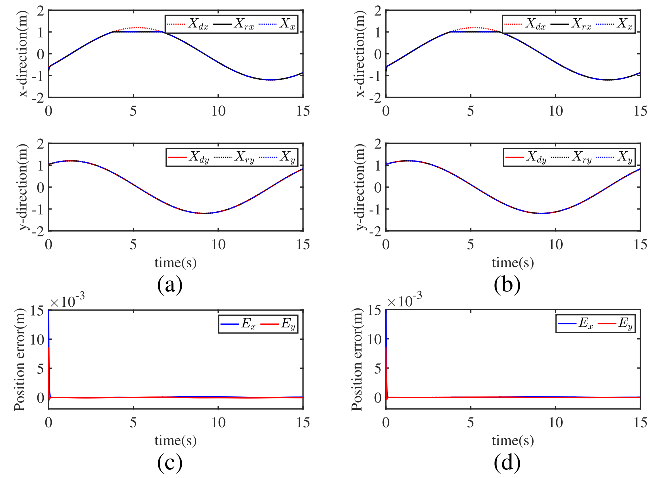


Fig. 2. Simulation results of position tracking. Position tracking of (a) ANIC and (b) ETANIC. Position error of (c) ANIC and (d) ETANIC.

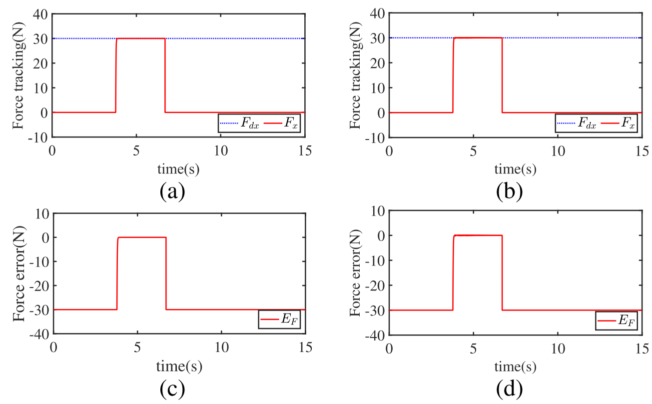


Fig. 3. Simulation results of force tracking. Force tracking of (a) ANIC and (b) ETANIC. Force error of (c) ANIC and (d) ETANIC.

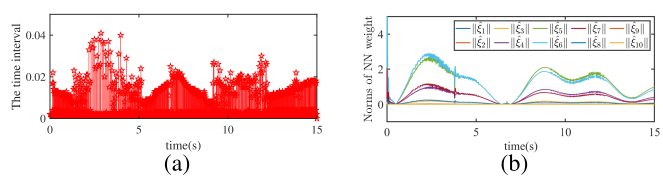


Fig. 4. Simulation results of ETANIC. (a) Time interval of the trigger. (b) Norms of NN weight.

TABLE II
RESULTS OF ETANIC WITH DIFFERENT μ IN SIMULATION

μ	0.1	0.3	0.5	0.7	0.9
RMS of $E_x(\times 10^{-3})$	6.72	6.93	6.93	6.94	6.97
RMS of $E_y(\times 10^{-3})$	2.55	2.51	2.51	2.51	2.52
RMS of E_F	1.37	1.36	1.35	1.38	1.40
Trigger rate(%)	40.3	37.5	35.3	34.6	33.6

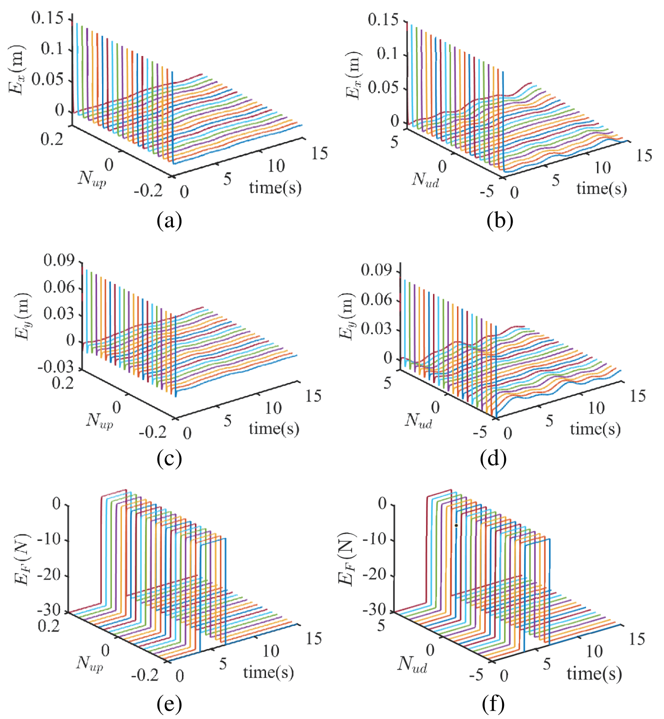


Fig. 5. Simulation results of ETANIC's robust test. (a) Position error in the X-axis with different model uncertainties. (b) Position error in the Y-axis with different disturbances. (c) Force error with different model uncertainties. (d) Position error in the X-axis with different disturbances. (e) Position error in the Y-axis with different model uncertainties. (f) Force error with different disturbances.

event and rms of E_F are relatively smaller, and the control performances of ETANIC are also relatively optimal.

2) Robust Test of ETANIC: For testifying the robustness of the proposed ETANIC scheme, the simulation studies are implemented by the robotic manipulator under different model uncertainties and disturbances, which are designed as $N_{\text{up}}Y_0$ and $N_{\text{ud}}d_1$, with $N_{\text{up}} = 0.02i - 0.2$ and $N_{\text{ud}} = 0.5i - 5$, $i = 0, 1, 2, \ldots, 20$. The value of parameter μ in (25) is set as 0.5 in this section. Fig. 5 shows the simulation results with different model uncertainties and disturbances. Fig. 5(a), (c), and (e) are the position tracking errors E_x and E_y and force tracking error E_F with different model uncertainties, respectively. Fig. 5(b), (d), and (f) are the position tracking errors E_x and E_y, and force tracking error E_F with different disturbances, respectively. The corresponding rms values of tracking errors under different model uncertainties and disturbances are demonstrated in Figs. 6 and 7, where the black horizontal line of each subfigure represents the corresponding rms values for tracking errors when $N_{\text{up}} = 0$ or $N_{\text{ud}} = 0$, respectively.

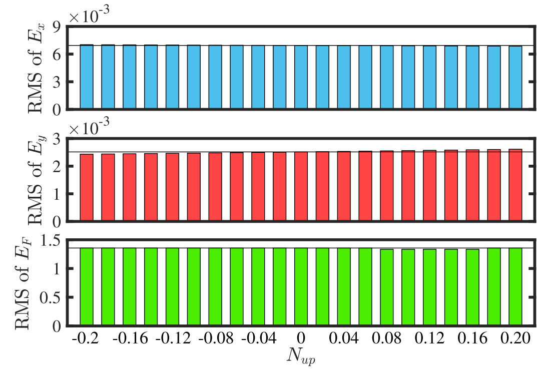


Fig. 6. RMS of tracking position errors with different model parameters.

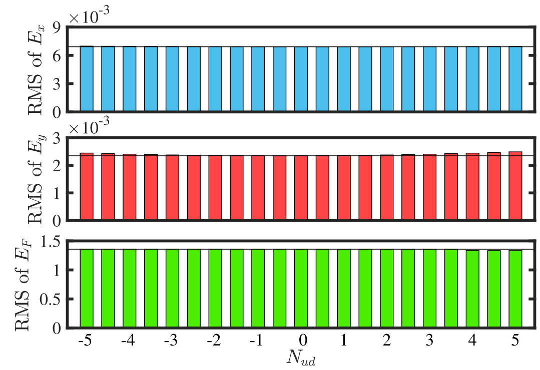


Fig. 7. RMS of tracking force error with different disturbances.

As shown in Fig. 5, the position and force tracking errors of the robotic system are UUB and convergent under different model uncertainties and disturbances. Figs. 6 and 7 illustrate that the rms of tracking errors remains near the corresponding horizontal lines, which indicates that the proposed ETANIC scheme can achieve the similar tracking performances for uncertain model uncertainties and disturbances. Based on the above analysis and results, the robotic system controlled by the proposed ETANIC scheme has robustness under different model uncertainties and disturbances.

B. Experimental Tests on a Virtual Robotic System

To further evaluate the performances of the proposed ETANIC strategy in the actual robotic system, we use the robot simulator CoppeliaSim (formerly known as V-REP) to conduct the experimental tests. The inertia of the robot, the physical effects of external forces, and the surrounding environment are taken into account by CoppeliaSim, which can well simulate the behavior of the robot in workspace. As exhibited in Fig. 8, the whole experimental setup includes the environment and a universal robot UR5, which is a six-degree-of-freedom (DOF) manipulator driven by six revolute joints, and the parameters of UR5 are given in Appendix B.

The UR5 robot can only be controlled by the joint position and velocity [51]. However, it is obvious that (18) and (20) are torque inputs, so it is necessary to transform the control signals into reference joint acceleration $\ddot{q}_r$ by the forward

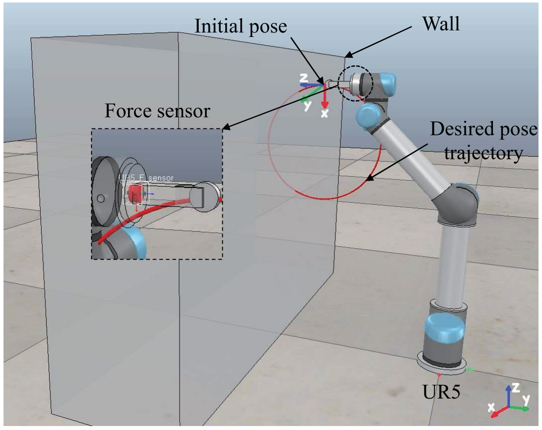


Fig. 8. Experimental setup.

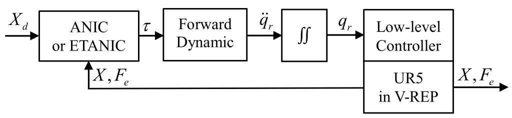


Fig. 9. System block diagram.

dynamics. Then, $\ddot{q}_r$ is integrated to obtain the reference joint configuration q_r, which will be then sent to the low-level controller of the robot manipulator, which is shown in Fig. 9.

The initial pose of the end-effector is $X(0) = [0.1091$ m, 0.45 m, 0.886 m, $\pi/2$ rad, 0 rad, and $-\pi/2$ rad$]^{\mathrm{T}}$. The desired pose trajectory is $X_d = [0.1091$ m, X_{dy}, X_{dz}, $\pi/2$ rad, 0 rad, and $-\pi/2$ rad$]^{\mathrm{T}}$, where X_{dy} and X_{dz} are described as (43). The end-effector will interact with a wall at $y = -0.55$ m and maintain the desired force $F_{dy} = 30$ N, where the contact force is measured by the force sensor, which is shown in Fig. 8. The sampling time and total time are 0.002 and 20 s, respectively,

$$\begin{cases} X_{dy} = 0.45 - 0.2\sin(0.5t) \text{ m} \\ X_{dz} = 0.686 + 0.2\cos(0.5t) \text{ m}. \end{cases} \tag{43}$$

The structures and parameters of RBFNN are set as $N_i = 18$, $N_m = 10$, $N_o = 6$, $\sigma_j = [-3, 3]_{10}$, $\delta_j = 2$, $\Gamma_1 = 0.01$, and $\eta = 10$. The parameters λ, K, μ, and α_0, β_0 of the controller and the impedance parameters are selected as

$$\begin{aligned} \lambda &= \mathrm{diag}([100, 100, 100, 25, 25, 25]) \\ K &= \mathrm{diag}([200, 200, 200, 50, 50, 50]) \\ \mu &= 0.5, \quad \alpha_0 = 0.05, \quad \beta_0 = 0.1, \quad \Lambda_d = 2I_{6\times 6} \\ D_d &= \mathrm{diag}([900, 800, 800, 150, 250, 250]) \\ K_d &= \mathrm{diag}([80, 200, 80, 10, 60, 20]). \end{aligned}$$

The compared results between ANIC and ETANIC on UR5 are demonstrated in Figs. 10 and 11. It is observed that the desired tracking performances can be achieved by ETANIC and ANIC, which corresponds to the results of numerical simulation in Section IV-A. Fig. 12(a) gives the changing of the trigger time interval with the experimental time under the event triggered conditions (25), and Fig. 12(b) gives the norms of NN weight in ETANIC. Due to the combination of ETC, the computation burden of ETANIC is significantly reduced than that of ANIC, which is shown in Fig. 12(a). And the norms of NN weight in ETANIC are bounded and the NN weight can converge to the optimal values as shown in Fig. 12(b). The rms values of tracking errors and trigger rates for different μ are demonstrated in Table III. As indicated from Table III, the control performances of ETANIC are also relatively optimal when $\mu = 0.5$. And in the experimental tests, ETANIC can still save more than 50% of the computational resources when comparing with ANIC.

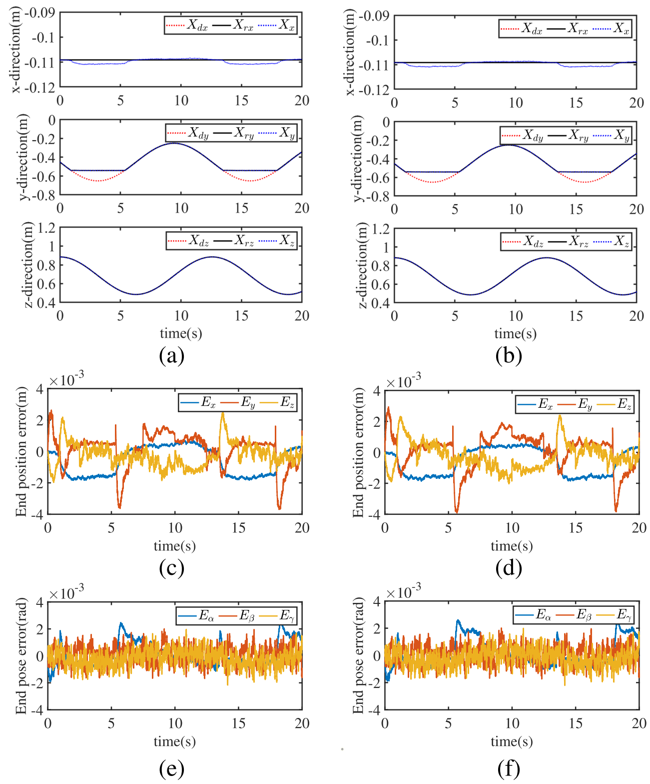


Fig. 10. Experimental results of position tracking. Position tracking of (a) ANIC and (b) ETANIC. Position error of (c) ANIC and (d) ETANIC. End pose error of (e) ANIC and (f) ETANIC.

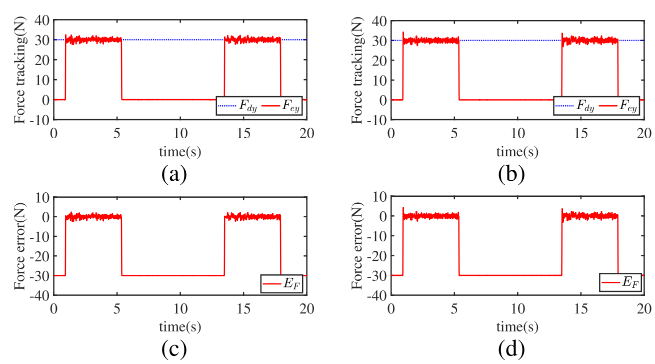
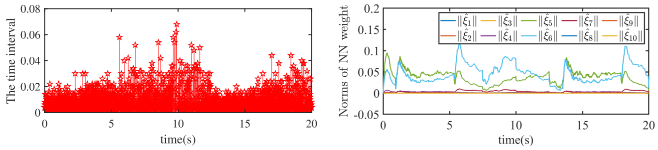


Fig. 11. Experimental results of force tracking. Force tracking of (a) ANIC and (b) ETANIC. Force error of (c) ANIC and (d) ETANIC.

C. Discussion on Simulations and Experimental Results

From Figs. 2, 3, 10, and 11, we can conclude that integrating ETC into the compliant control framework will not descend the tracking performances of the robotic system. Both ANIC and the proposed ETANIC can be implemented on the desired



(a) (b)

Fig. 12. Experimental results of ETANIC. (a) Time interval of the trigger. (b) Norms of NN weight.

TABLE III
RESULTS OF ETANIC WITH DIFFERENT μ IN EXPERIMENT

μ	0.1	0.3	0.5	0.7	0.9
RMS of $E_x(\times 10^{-4})$	9.88	9.83	9.89	9.94	10.02
RMS of $E_y(\times 10^{-3})$	1.22	1.16	1.04	1.21	1.24
RMS of $E_z(\times 10^{-4})$	9.36	6.78	6.86	9.64	9.81
RMS of E_F	1.94	1.93	1.93	1.94	1.95
Trigger rate(%)	59.4	53.0	44.8	42.9	39.7

control objective in the whole space. Meanwhile, due to ETC, the computational burden of the system is reduced, which is shown in Figs. 4(a) and 12(a). Compared with the results of ANIC, the proposed ETANIC can achieve the objective of the stable position tracking and force tracking with less computational burden and communication cost. According to Theorem 2, it can be concluded that under different values of μ, event-triggered conditions (25) will lead to similar tracking performances, which is also verified in Tables II and III. The simulation results under different model uncertainties and disturbances show that the proposed ETANIC scheme has the robustness, which is shown in Figs. 5–7.

Moreover, according to the results in Tables II and III, it can be found that the event trigger rates are different under the same μ, the reason being different robotic systems used in numerical simulation and experimental tests. Compared with the two-rigid-link robotic manipulator in Section IV-A, the UR5 robot in Section IV-B is a six-DOF manipulator, where the tracking errors including position, pose, and force errors in the Cartesian space make the event-triggered conditions easier to be satisfied and thus lead to high trigger rates.

V. CONCLUSION

The ETANIC scheme is presented for the robotic manipulators with uncertainties in this article. Using the Lyapunov stability theory, the event-triggered conditions and update law for RBFNN are designed, respectively, to avoid the Zeno behavior and approximate the model uncertainties and disturbances while analyzing and proving the stability of the whole control system. The results of simulation tests show that the robotic system controlled by the proposed ETANIC scheme has the robustness under different model uncertainties and disturbances. The comparative experiments between ANIC with constant sampling and ETANIC with event-based sampling are presented, and the control performances of ETANIC under the event-triggered conditions with different parameters are analyzed. The results show that ETANIC can reduce the computation cost of the robotic systems while ensuring control performances. In the future, the proposed method will be extended to the collaboration and compliant control of multiple robotic manipulators and human–robot interaction and collaboration.

TABLE IV
PARAMETERS OF THE UR5 ROBOT

Link	1	2	3	4	5	6
q(rad)	q_1	q_2	q_3	q_4	q_5	q_6
d(m)	0.08946	0	0	0.1092	0.09465	0.1723
a(m)	0	-0.425	-0.3923	0	0	0
α(rad)	$\pi/2$	0	0	$\pi/2$	-$\pi/2$	0
Mass(kg)	3.700	8.393	2.330	1.219	1.219	0.1897
L_c(m)	L_{c1}	L_{c2}	L_{c3}	L_{c4}	L_{c5}	L_{c6}

where q, d, a and α are the D-H parameters, and L_c is the center of mass, $L_{c1} = [0, -0.02561, 0.00193]$, $L_{c2} = [0.2125, 0, 0.1134]$, $L_{c3} = [0.15, 0, 0.0265]$, $L_{c4} = [0, -0.0018, 0.01634]$, $L_{c5} = [0, -0.0018, 0.01634]$, $L_{c6} = [0, 0, -0.001159]$.

APPENDIX

A. Dynamics of Two-Rigid-Link Robotic Manipulator

The dynamics of the two-rigid-link robotic manipulator used in Section IV-A is given by [52]

$$M(q) = \begin{bmatrix} a_1 + a_2 + 2a_3c_2 + I_1 + I_2 & a_2 + a_3c_2 + I_2 \\ a_2 + a_3c_2 + I_2 & a_2 + I_2 \end{bmatrix}$$

$$C(q,\dot{q}) = \begin{bmatrix} a_3\dot{q}_2 & a_3(\dot{q}_1 + \dot{q}_2) \\ a_3\dot{q}_1 & 0 \end{bmatrix}, \quad G(q) = \begin{bmatrix} a_4c_1 + a_5c_{12} \\ a_5c_{12} \end{bmatrix}$$

where $a_1 = m_1l_{c1}^2 + m_2l_1^2$, $a_2 = m_2l_{c2}^2$, $a_3 = -m_2l_1l_{c2}s_2$, $a_4 = (m_1l_{c1} + m_2l_1)g$, and $a_5 = m_2l_{c2}g$. I_i, m_i, l_i, and l_{ci} denote the moment of inertia, mass, length, and centers of mass of link i, respectively, $i = 1, 2$. $g = 9.8$ m/s^2 is the acceleration of gravity. s_i, c_i, s_{ij}, and c_{ij} denote $\cos(q_i)$, $\sin(q_i)$, $\sin(q_i + q_j)$, and $\cos(q_i + q_j)$, respectively, $i = 1, 2$ and $j = 1, 2$.

The Jacobian matrix can be described as

$$J(q) = \begin{bmatrix} -l_1s_1 - l_2s_{12} & -l_2s_{12} \\ l_1c_1 + l_2c_{12} & l_2c_{12} \end{bmatrix}.$$

B. Parameters of the UR5 Robot

The nominal values of parameters of the UR5 robot used in Section IV-B are given in Table IV.

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